

Offline knowledge-base construction

Clinical corpus
(guidelines, PubMed/PMC, MedQA textbooks, drug refs)

Passage chunking
2,294,038 chunks (~512 tokens)

MedCPT encoder + FAISS IVF-PQ
(2.29M vectors)

UMLS negation/temporal extraction
+ Kuzu property graph
(2.50M nodes; 632,931
NEGATES / TEMPORAL edges)

Online inference (fully offline device)

Clinical query

Stage 1 · Query rewriting
Qwen2.5-7B, T=0.0
<=30-word canonical form

Stage 2 · Dense retrieval
(MedCPT + FAISS)

Lexical retrieval
(BM25)

Stage 3 · Graph expansion
(Kuzu, <=2 hops)

RRF fusion +
relation-aware rerank
(gamma = 0.15)

Top-10 evidence
passages

Stage 4 · Beta-gated verification

Evidence faithfulness
(NLI phi)

Graph consistency
(S_g)

$$V = \beta \phi + (1 - \beta) S_g \geq \tau^* \quad (\text{WAR} \leq 10\%)$$

$$V < \tau^*$$

$$V \geq \tau^*$$

Structured refusal +
knowledge-gap card (F2)

Stage 5 · Evidence-grounded generation
(Qwen2.5-7B Q4_K_M, single-flight GPU lock)

Verified clinical response [E#]
+/- F1 cross-modal alert
-> human review

offline / online boundary